

Serious Shea: Business Plan

August 2021

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1. INTRODUCTION

Executive Summary

ABOUT THE BUSINESS

- Shea processing is a major industry in Burkina Faso; however, it still involves outdated methods: depleting forest populations, emitting significant carbon emissions, causing a number of health and hygiene risks, and doing little to improve the socio-economic situation in the country (as the real value is often added during processing outside of Burkina Faso)
- The Serious Shea business aims to transform the shea sector in Burkina Faso by implementing collection sites and processing centres **in local communities**, where the training and technology is provided to process shea butter cleanly, efficiently and sustainably
- The business will ensure women's groups, who are predominantly responsible for the processing, are able to produce the shea butter more efficiently and also achieve significantly better access-to-market (receiving ~1.5x higher prices). It will also use clean technology and drive reforestation / afforestation around each centre and collection site, generating certified carbon credits by decarbonising the process and REDD+

FUNDING

- The business has received **>US\$0.5m** grant funding from the **African Development Bank** to conduct preparation activities since 2017, including a pilot operation in Ouagadougou, detailed engineering designs, processing tests, mapping, operational plans, logistics, country-wide surveys, marketing, etc.
- Serious Shea requires **US\$10m** from the DFC in order to commence Phase 1 in 2021:
 1. Construct 8 x collection sites (starting in Didyr, Sanguié Province and Djigologo, Loba Province) and 4 x processing centre (starting in Koudougou, Boulkiemdé Province) in 2021, with production commencing in 2022
 2. Collect ~6,000mt of dried kernels, and produce and export ~2,400mt of shea butter in 2022 (Revenue: >US\$9m)
 3. Create direct employment for ~1,280 staff (90% women) & benefit 10,000s of women processors by providing ~1.5x higher prices plus 10% equity

COUNTRY-WIDE IMPLEMENTATION

- Following Phase 1, Serious Shea plans to roll-out at a regional scale, operating multiple collection sites and large-scale processing centres throughout Burkina Faso and Ghana by 2024

1. INTRODUCTION

Understand shea butter processing

Fruit



- Moisture content of 75-80%
- Raw, unprocessed shea fruit, prior to de-pulping

Nut



- Moisture content of 30-40% as fresh, <6-8% as processed
- De-pulped shea fruit that has been dried

Kernel



- Moisture content of 30-40% as fresh; <6-8% as processed
- Dried shea nut that has been boiled, deshelled and dried

Butter



- Moisture content of <6%
- Shea kernel that has been processed into refined butter, for sale by Serious Shea to its end market

Note: The pulp forms from 33-80% of the weight of the whole fruit, which is typically between 20-30 g.

Weight and moisture content vary of each product varies as a result of processing modalities. There are 1) the natural characteristic and 2) the characteristic after processing linked to targeted quality. These parameters vary from group to group.

1. INTRODUCTION

Business model

Collection sites:

- Semi-mechanised sites used by women's groups to process fruit into shea kernels, supervised by a Serious Shea employees
- Sites will be located in (i) PIF areas (Programme d'Investissement Forestier), (ii) areas with high levels of shea fruit yields and (iii) areas with sufficient women's groups eager to partner with Serious Shea
- Utilise clean technology and renewable energy, but retain the traditional process for women's groups to operate easily
- Serious Shea assumes that each women processor collects 250kg to 500kg of kernels each season (in reality, this assumption is conservative given that many women collect 20kg per day x 90 day season = 1,800kg of fruit)
- Number and size of collection sites depends on distance women can transport fruit, and the requirements of the processing plant
- Dried shea kernels purchased by Serious Shea (6,000mt in 2021/22) at approx. 1.5x higher prices from women's groups, and processed into shea butter
- Each collection site will also produce 300t / site / year of compost from the unused fruit meat

Processing centres:

- The Phase 1 plants will be built in 2021 (starting in Koudougou), each processing 1,500mt of shea kernels into 600mt / year of unrefined organic shea butter in the 2021/22 season
- Operated by Serious Shea employees (women) only
- Exported to international cosmetic buyers initially and then food buyers
- 10% of profit provided back to participating women's groups

1. INTRODUCTION

Current situation: Inefficient and labour-intensive processing methods

1. Collection & Boiling

Shea nuts and fuelwood are harvested and carried for miles, driving widespread deforestation, before inefficient 3-stone fires are used to boil the nuts



2. Drying & Crushing

The kernels are dried and hand-crushed, requiring long days of heavy physical labour and significant negative health impacts (e.g., smoke inhalation)



3. Cooking & Extraction

The women's groups use their hands and reused metal and plastic containers to complete the final extraction process, with limited hygiene measures



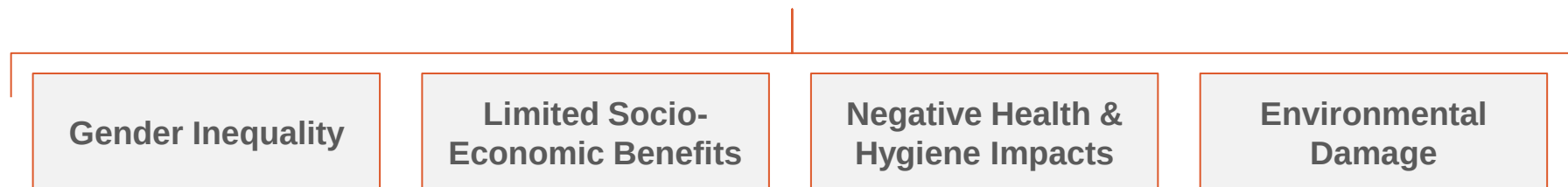
- The existing butter-making process in Burkina Faso drives significant deforestation and greenhouse gas emissions and requires women's groups to conduct heavy physical labour. Furthermore, the lack of access-to-market means that the women's groups do not receive appropriate remuneration for their inputs and failing to improve their socio-economic status despite their hard labour.

1. INTRODUCTION

Current situation: Significant deforestation and poor prices

- The shea sector is the 3rd largest export sector in Burkina Faso, where the production of shea products involves nearly three million people in Burkina Faso, required for picking, preparation, and processing. This production is often conducted by women's groups and helps to improve the living conditions of rural households
- However, the methods for the production of shea almonds and butter involves significant wood consumption - approximately 20kg of wood per kg of butter at farm. This has serious repercussions on agricultural productivity, natural habitats, climate change and local health
- Furthermore, local shea tree populations are plummeting due to the demand for the nuts and fuelwood. In Burkina Faso, between 1990 and 2010, the total forest area has decreased by 68,470 km² – 56,490 km², representing a loss of 17.5%
- Finally, due to limited access-to-market, the women's groups achieve low prices for the butter from local traders, forcing them to harvest and process the butter on a daily basis

Impact of Current Production



Our solution: Shea butter eco-processing centres & access-to-market

- Serious is driving an effort to transform the lives of rural women shea workers in Burkina Faso through the development of multiple collection sites and 3 x large processing centres:
 - *Each collection site will be fully equipped to process shea fruit into shea kernels*
 - *Each processing centre will be fully equipped to process shea kernels into shea butter, with further fractionization and refining potentially*
 - *The facilities will draw on renewable energy sources (e.g., biogas and solar PV) and include all the equipment necessary for clean, efficient, and environmentally-friendly shea butter production*
- More specifically, each centre to be equipped with an optimal mix of the following:
 - *Solar cookers and steam boilers to dry and boil the nuts cleanly and efficiently*
 - *Solar-powered motors, which enable the women's groups to produce the shea paste quickly without the strenuous physical labour of traditional manual methods*
 - *Large-scale improved cookstoves to boil the mixture and extract the butter*
 - *All the necessary apparatuses and equipment to store, transport, package and export the final shea butter product*

1. INTRODUCTION

Our solution: Renewable energy, mechanical processing, and fair prices

1. Collection & Boiling

Solar cookers and steam boilers are used to process the shea nuts, eliminating the need for inefficient traditional wood-burning alternatives



2. Drying & Crushing

Mechanical crushing and milling equipment significantly reduce the physical burden on the women's groups and increase their productivity



3. Cooking & Extraction

Gas stoves, butter extraction apparatuses and food-grade packaging equipment prepare the shea butter for export



- Serious Shea's sustainable approach to shea butter production will transform the shea industry in Burkina Faso, dramatically reducing deforestation, GHG emissions and various negative health impacts through the use of clean technology solutions, and improve the livelihoods of women's groups across the country through higher productivity and improved access-to-market (~1.5x increase in income)
- The centres will **not** burn any wood!

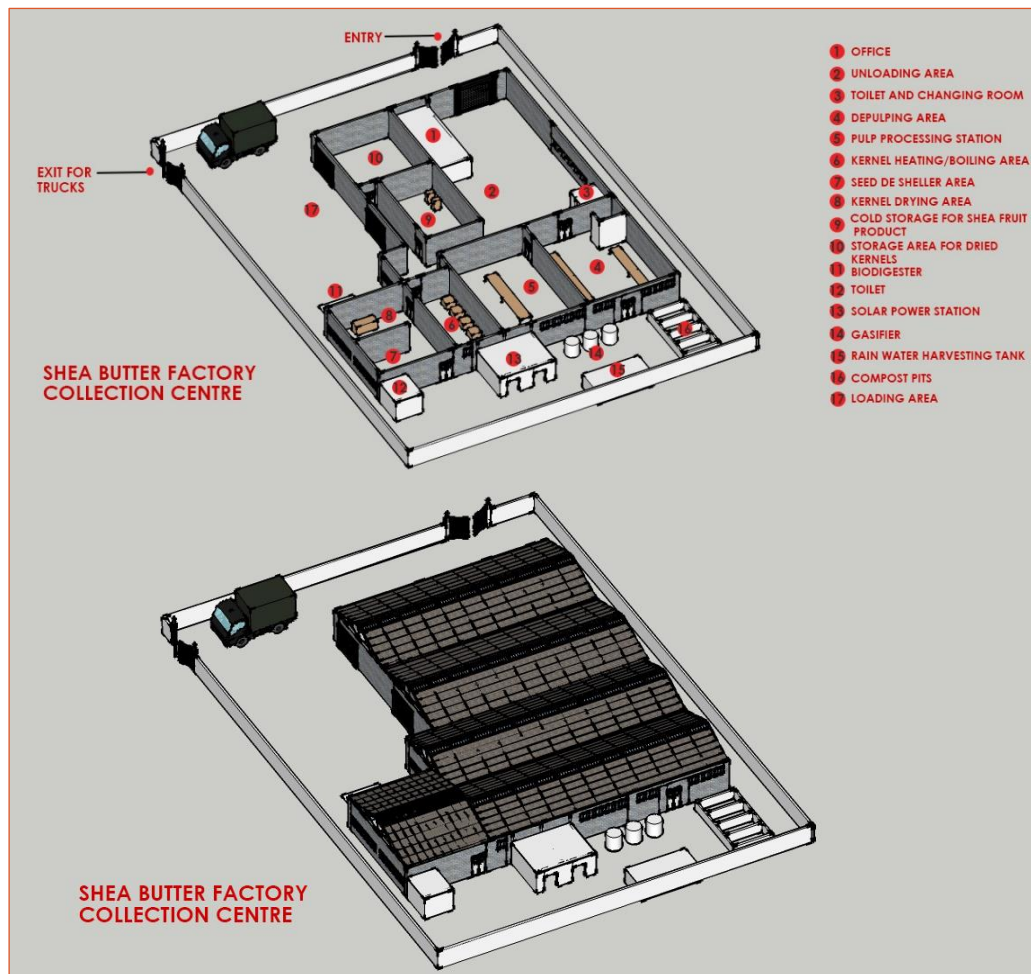
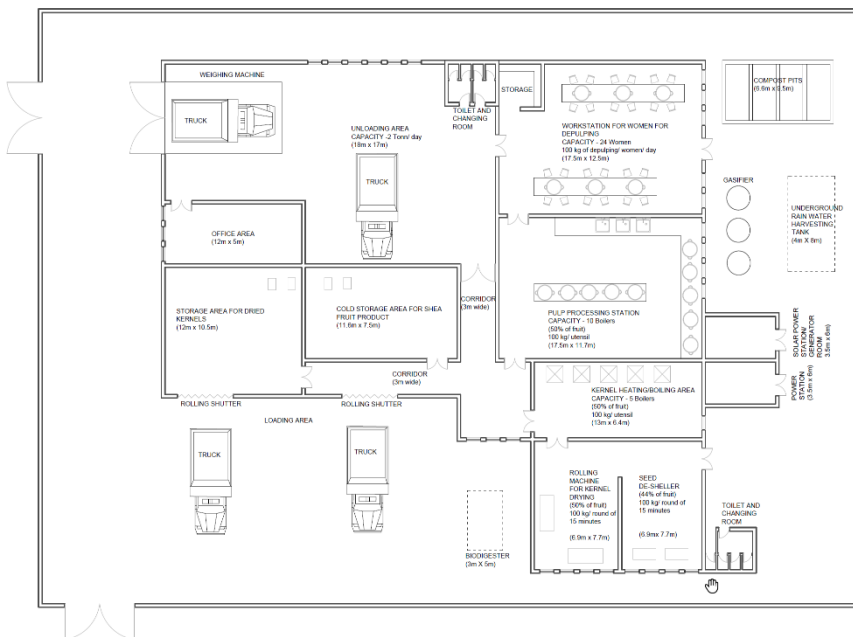
1. INTRODUCTION

Our solution: Renewable energy, mechanical processing, and fair prices

Collection Sites

- Each set of collection sites will be fully equipped to process shea fruit into shea kernels
- The facilities will draw on renewable energy sources (e.g., biogas and solar PV) and include all the equipment necessary for clean, efficient, and environmentally-friendly shea butter production

SHEA BUTTER COLLECTION UNIT LAYOUT - OPTION 02
CAPACITY - 2 TONNI/ DAY



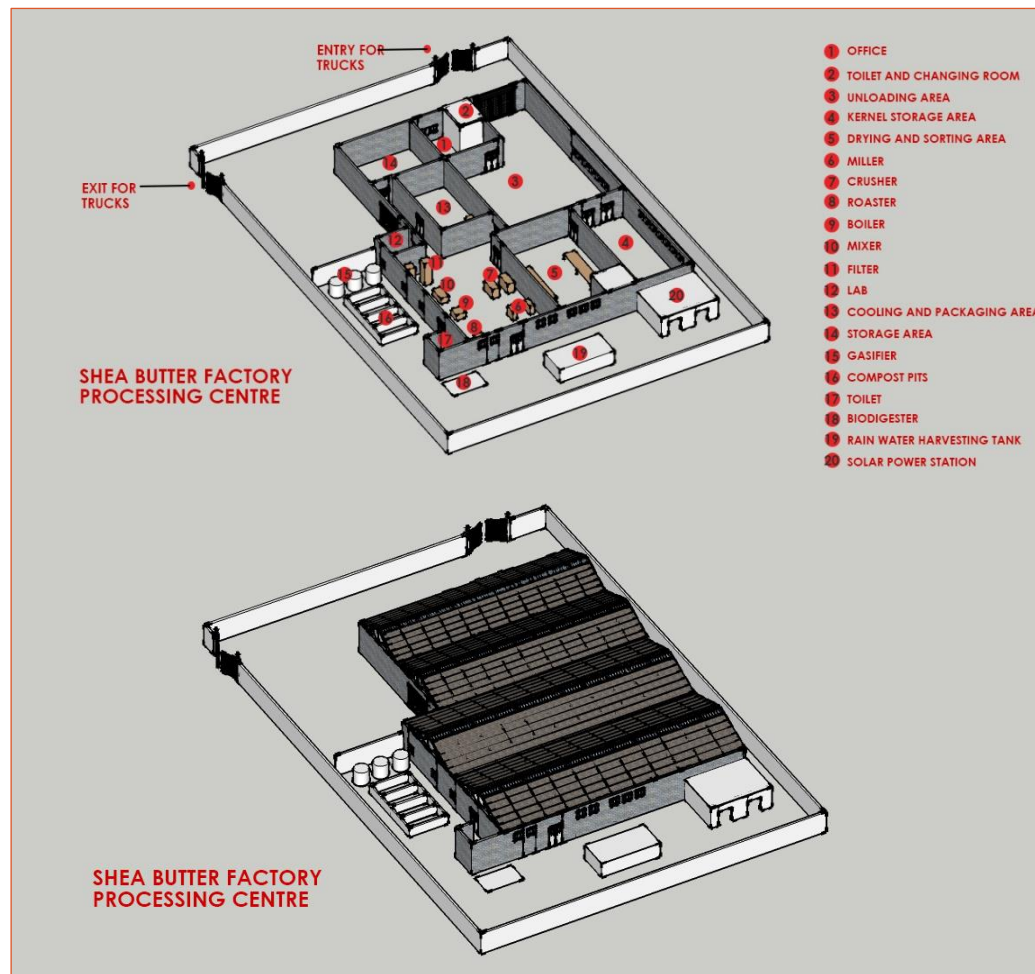
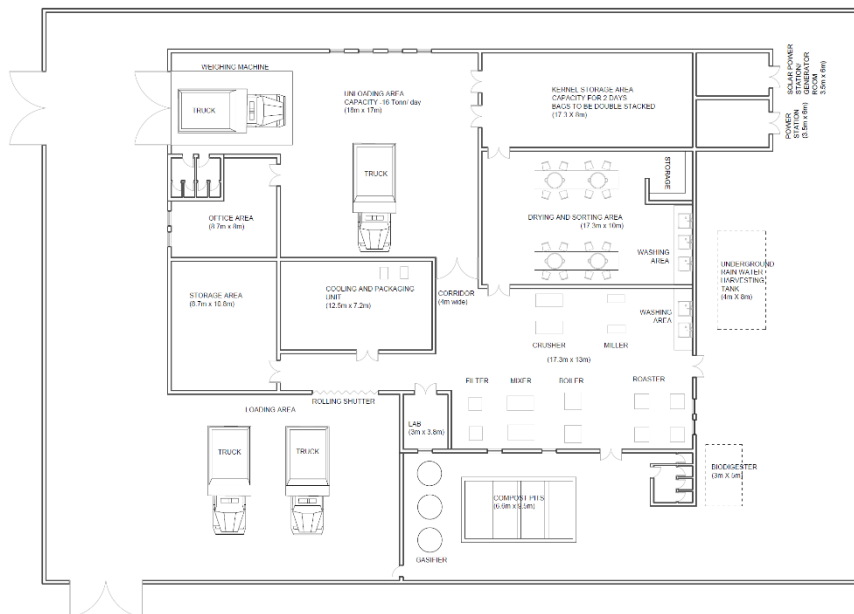
1. INTRODUCTION

Our solution: Renewable energy, mechanical processing, and fair prices

Processing Centres

- Each processing centre will be fully equipped to process shea kernels into shea butter, with further fractionization and refining potentially
- The facilities will draw on renewable energy sources (e.g., biogas and solar PV) and include all the equipment necessary for clean, efficient, and environmentally-friendly shea butter production

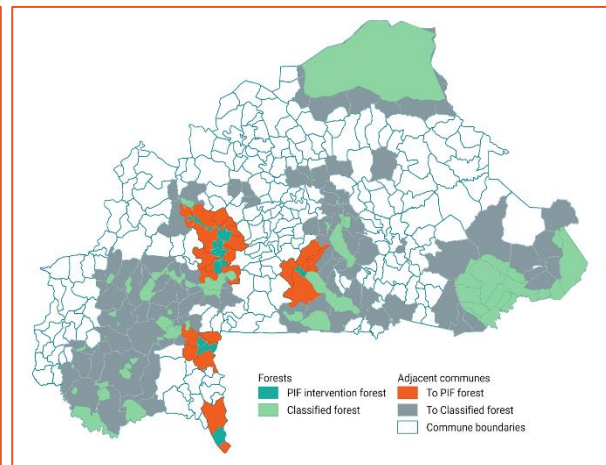
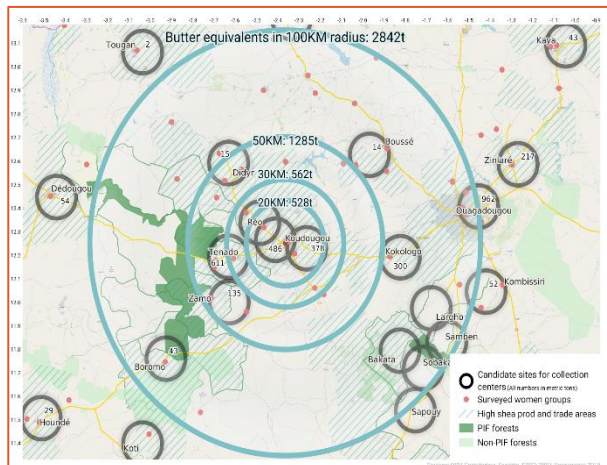
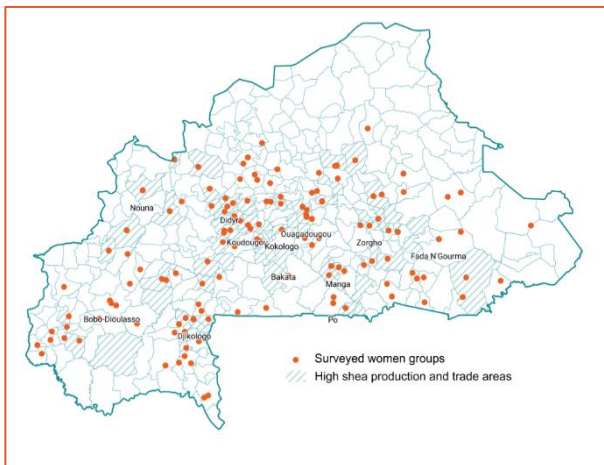
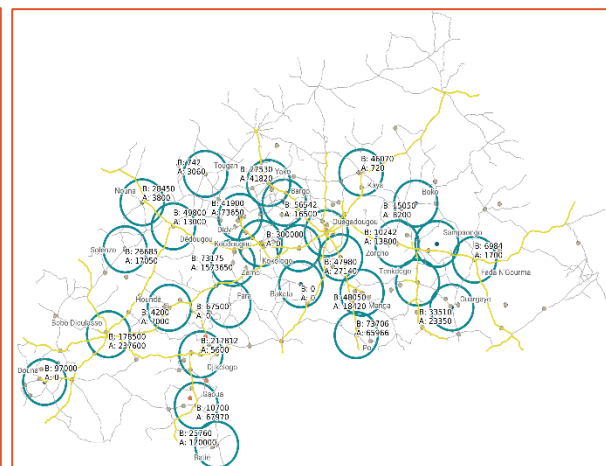
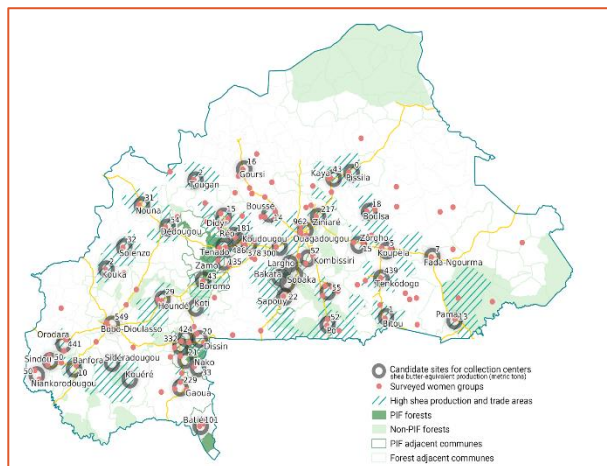
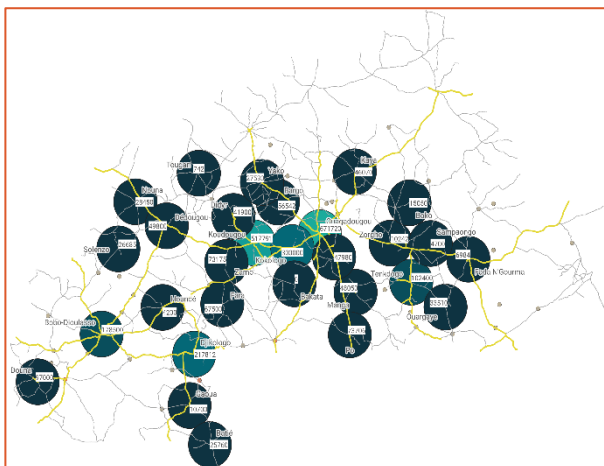
SHEA BUTTER PROCESSING UNIT LAYOUT
CAPACITY - 16 TONNI/ DAY



1. INTRODUCTION

Locations: Identified optimal locations for collection sites and processing centres

- Serious Shea has been working with Sensonomic to conduct sophisticated mapping and ground surveys, identifying optimal sites



1. INTRODUCTION

Women Processors: Serious Shea has identified and partnered with >4,000 women

- Serious Shea has conducted surveys across Burkina Faso to identify and partner with women processors. These women will use the collection sites to process the fruit into dried kernels (using safe, clean and wood-free technology) and then sell the dried kernels to Serious Shea (receiving 1.5x higher income and 10% of the annual profits) – extract from dataset below:

Group Name	Surname	First Name	Region	Province	Commune	Village	Geographic Scope	# of Members	Dried Kernels Identified by Serious Shea (sold as kernels)			Dried Kernels Identified by Serious Shea (currently processed into butter)		
									2017	2018	2019	2017	2018	2019
									Kilograms of Dried Kernels			Kilograms of Dried Kernels		
Niorognè	Kandolo	Fanta	Centre-Ouest	Sanguie	Didyr	Goumi	Communale	30	600.00	360.00	240.00	75.00	37.50	15.00
Goulou Nyè	Kantoro	Eya	Centre-Ouest	Sanguie	Didyr	Sodjir	Communale	55	1,200.00	600.00	360.00	93.75	37.50	15.00
Bakondo De Doolo	Kankouan	Edo	Centre-Ouest	Sanguie	Didyr	Didyr	Communale	31	4,200.00	3,000.00	1,440.00	150.00	75.00	30.00
Korotou De Nidiolo N'Tounou	Kanzie	Edama	Centre-Ouest	Sanguie	Didyr	N'Tounou	Villageoise	53	9,000.00	7,200.00	3,600.00	187.50	112.50	56.25
Koroto	Dakani	Ejoin	Centre-Ouest	Sanguie	Didyr	Ladiou	Villageoise	36	600.00	360.00	1,200.00	22.50	15.00	30.00
Sono Nyè	Bazie	Genevieve	Centre-Ouest	Sanguie	Didyr	Mogaya	Villageoise	54	9,000.00	9,000.00	6,000.00	150.00	123.75	112.50
Kyidone Nyè	Kanko	Mamounata	Centre-Ouest	Sanguie	Didyr	Zhiladyr	Communale	38	720.00	480.00	960.00	375.00	112.50	93.75
Ça-Dwa-Nyè/Koroto	Kantoro	Sylvie	Centre-Ouest	Sanguie	Didyr	Kwen	Communale	20	12,000.00	7,200.00	9,600.00	225.00	150.00	187.50
Pougnè	Dango	Sallimata	Centre-Ouest	Sanguie	Didyr	Nedjendji(Pouini)	Communale	32	3,960.00	2,400.00	3,000.00	112.50	75.00	93.75
Soulougè	Kando	Sito Rachelle	Centre-Ouest	Sanguie	Didyr	Mousseo	Communale	26	2,400.00	1,800.00	960.00	112.50	45.00	75.00
Kinègnè	Kankouan	Ezhilboué	Centre-Ouest	Sanguie	Didyr	Eshinadyr	Communale	15	4,560.00	5,040.00	6,000.00	187.50	150.00	187.50
Koroto	Kanzie	Kéfol	Centre-Ouest	Sanguie	Didyr	Bissou	Communale	33	1,800.00	2,280.00	360.00	37.50	-	37.50
Yichipiè	Kanko	Efoma	Centre-Ouest	Sanguie	Godyr	Demapoun	Communale	34	8,400.00	8,400.00	10,800.00	300.00	168.75	300.00
Club Des Paysannes Forestières	Bado	Clarisse	Centre-Ouest	Sanguie	Godyr	Godyr	Communale	41	4,080.00	6,600.00	6,600.00	112.50	168.75	112.50
Yilwalgnè	Bazie	Forzia	Centre-Ouest	Sanguie	Godyr	Yorkio-Nediolo	Communale	24	3,120.00	3,120.00	4,200.00	150.00	-	150.00
Sulagnè	Kando	Eguilboué	Centre-Ouest	Sanguie	Godyr	Semaga	Communale	36	2,760.00	3,720.00	3,720.00	131.25	75.00	131.25
Koroto De Boho	Kanzie	Ellili	Centre-Ouest	Sanguie	Godyr	Boho	Communale	27	2,280.00	1,440.00	2,400.00	225.00	150.00	225.00
Gnidwignè	Kogo	Mariam	Centre-Ouest	Sanguie	Didyr	Pouini-Nord	Communale	48	9,600.00	11,040.00	11,040.00	225.00	225.00	225.00
Yiwèngnègnènyè	Kanko	Eyimbé	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	35	8,400.00	8,400.00	9,600.00	168.75	168.75	112.50
Sonognè	Kanzongo	Kézoumbié	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	47	11,280.00	11,280.00	11,280.00	112.50	112.50	187.50
Gulagnè	Kanzongo	Keyom	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	42	10,080.00	7,200.00	12,000.00	300.00	150.00	300.00
Turnegnè	Kanko	Kédjima	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	55	26,400.00	22,320.00	28,800.00	112.50	75.00	112.50
Yiwè	Bayala	Malmouna	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	60	21,600.00	9,840.00	31,200.00	168.75	168.75	168.75
Sulagnè	Kansole	Véronique	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	36	8,400.00	6,000.00	10,800.00	150.00	150.00	150.00
Ozado	Kientega	Képima	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	15	3,600.00	4,800.00	3,600.00	112.50	112.50	112.50
Sanadoné	Kanzongo	Kézoum	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	43	10,320.00	7,440.00	14,640.00	225.00	112.50	225.00
Sonognè	Kanyili	Képoulou	Centre-Ouest	Sanguie	Kordie	Kordie	Communale	33	7,920.00	6,240.00	10,080.00	150.00	150.00	150.00
Sandonegnè	Yili	Erona	Centre-Ouest	Sanguie	Kordie	Ninon	Communale	55	19,800.00	20,400.00	24,000.00	225.00	150.00	225.00
Gulagnè De Kiro	Kansinga	Elisabeth	Centre-Ouest	Sanguie	Kordie	Kiro	Communale	55	19,800.00	16,800.00	22,800.00	300.00	150.00	300.00
Kidonemye De Goumi	Kando	Eguiné	Centre-Ouest	Sanguie	Didyr	Goumi	Communale	23	600.00	360.00	240.00	93.75	93.75	18.75
Kilounye De Kwen	Kansane	Elzilibé	Centre-Ouest	Sanguie	Didyr	Kwen	Communale	30	960.00	600.00	1,080.00	75.00	45.00	75.00
Yiwéazhena De Yierdion	Kansane	Jeanne	Centre-Ouest	Sanguie	Didyr	Yéridion	Provinciale	30	12,000.00	7,200.00	14,400.00	7,500.00	3,750.00	5,625.00
Pynognin De Ziladié	Kankouan	Damata	Centre-Ouest	Sanguie	Didyr	Ziladié	Communale	32	840.00	600.00	1,200.00	150.00	75.00	131.25
Zouinbilou	Kawar	Ellie	Centre-Ouest	Sanguie	Didyr	Pouini	Communale	60	-	1,440.00	1,800.00	-	-	-
Soulou Nye De Tonton	Kandolo	Eyimbé	Centre-Ouest	Sanguie	Didyr	Tonton	Communale	33	1,080.00	960.00	1,200.00	225.00	112.50	300.00
Kiédonéyin	Bado	Germaine	Centre-Ouest	Sanguie	Didyr	Mousoumou	Communale	86	1,200.00	600.00	960.00	75.00	45.00	45.00
Kidonié	Kanko	Zenabo	Centre-Ouest	Sanguie	Didyr	Kwen	Communale	29	600.00	240.00	600.00	-	-	-
Cedwaneneyen	Kanzole	Epiman	Centre-Ouest	Sanguie	Didyr	Moussoumou	Communale	30	-	1,680.00	2,640.00	562.50	375.00	750.00
Sulune	Kantoro	Marguerite	Centre-Ouest	Sanguie	Didyr	Moussoumou	Communale	34	1,800.00	1,200.00	2,160.00	112.50	168.75	187.50
Dye Edu T'A Pwa Mun	Kanzongo	Elou	Centre-Ouest	Sanguie	Didyr	Moussoumou	Communale	36	2,160.00	1,560.00	2,400.00	60.00	37.50	93.75
Gnidougnin	Badiel	Catherine	Centre-Ouest	Sanguie	Didyr	Moussoumou	Communale	30	960.00	840.00	1,800.00	75.00	60.00	75.00
Wourouandouan	Kanzongo	Edo	Centre-Ouest	Sanguie	Didyr	Didyr	Communale	50	3,000.00	2,400.00	3,120.00	225.00	281.25	375.00
Kelaigmen	Kandiel	Damata	Centre-Ouest	Sanguie	Dassa	Nebia	Communale	47	1,440.00	1,920.00	3,600.00	-	198.75	225.00
Nidoumoukoka	Bazie	Marguerite	Centre-Ouest	Sanguie	Didyr	Pouini	Village	30	10,800.00	12,000.00	14,400.00	262.50	375.00	375.00
Sulanyen	Kanzie	Zita	Centre-Ouest	Sanguie	Didyr	Pouini	Village	45	3,600.00	3,000.00	5,520.00	-	-	-
Sabonanyen	Kanko	Jacheline	Centre-Ouest	Sanguie	Didyr	Pouini	Village	35	4,200.00	3,960.00	6,000.00	-	-	-
Ogoumè	Kankouan	Moussoumou	Centre-Ouest	Sanguie	Dassa	DSS	Communale	45	14,400.00	12,000.00	18,000.00	-	-	-

2. THE TEAM

The team includes local and international shea and agribusiness experts

Serious Shea Management Team



William Kwende (Burkinabe). Highly experienced agribusiness developer. William specialises in establishing large-scale renewable energy and agricultural projects in rural west Africa. He seamlessly integrates with the highest levels in government and the lowest levels in local communities.



Jon White (British). CEO and co-founder of ImpactAgri. More than 25 years experience managing companies and operations in Africa, Europe and the USA. Broad experience working with large input suppliers, farming companies, retailers, off takers, government and investment community.



Felicite Yameogo (Burkinabe). Key sustainable production representative at the European Commission, and recognised across west Africa as a leading figure in sustainable agribusiness processing. Responsible for training 10,000s of west Africa women in shea butter processing.



Alastair Woods (South African). Experienced partner at ImpactAgri with strong consulting experience working with multinational project developers. Alastair has spent 10 years working in a number of countries across sub-Saharan Africa to develop agricultural (e.g., blueberries, figs, tree nuts, etc.) and renewable energy projects.

Institutional Support



The **Burkina Faso government**, represented by the Ministry of Environment, is strongly supportive of Serious Shea



The **AfDB** is supporting Serious Shea by funding the preparation activities, including forestry mapping, technology designs and site feasibility assessments



The **World Bank** has provided grant funding to Serious Shea and continues to help drive the implementation of the project in Burkina Faso

2. THE MARKET

Market: Understanding the potential buyers is key

PERSONAL CARE AND COSMETICS SECTOR

- Cosmetics sector is < 20% of the shea butter industry but values the strong ESG performance of Serious Shea
- Serious Shea will sell 1,000 - 2,000 tonnes per annum of unrefined, refined and organic shea butter
 - Through its direct connection with buyers in the sector, it could secure a better price than if shea sold directly by local groups
- This is based on securing contracts with:
 - 2 - 4 of the larger companies procuring around 300 - 500 tonnes each per annum
 - 10 - 20 smaller companies procuring between 10 - 50 tonnes each per annum
 - Our network within the industry and the customer database has been developed via trade shows, relationship building and presence in the market

CONFECTIONARY AND FOOD SECTOR

- The confectionary and food sector accounts for 80-90% of the Burkina Faso Shea butter industry
- A large volume of shea-derived products are provided by AAK, Loders IOI and others who procure nuts and are vertically integrated
- Serious Shea has the option to supply companies with fractionation plants such as ISF, Wilmar, Fuji, Musim Mas, Sime Darby
 - While they purchase up to 50,000 tonnes / year each (refined or unrefined), the volumes sold will depend on price and logistics
 - Volumes sold could sell 1 - 5,000 tonnes of shea butter each
- To secure a higher margin, Serious Shea could explore getting more involved in production of shea stearin

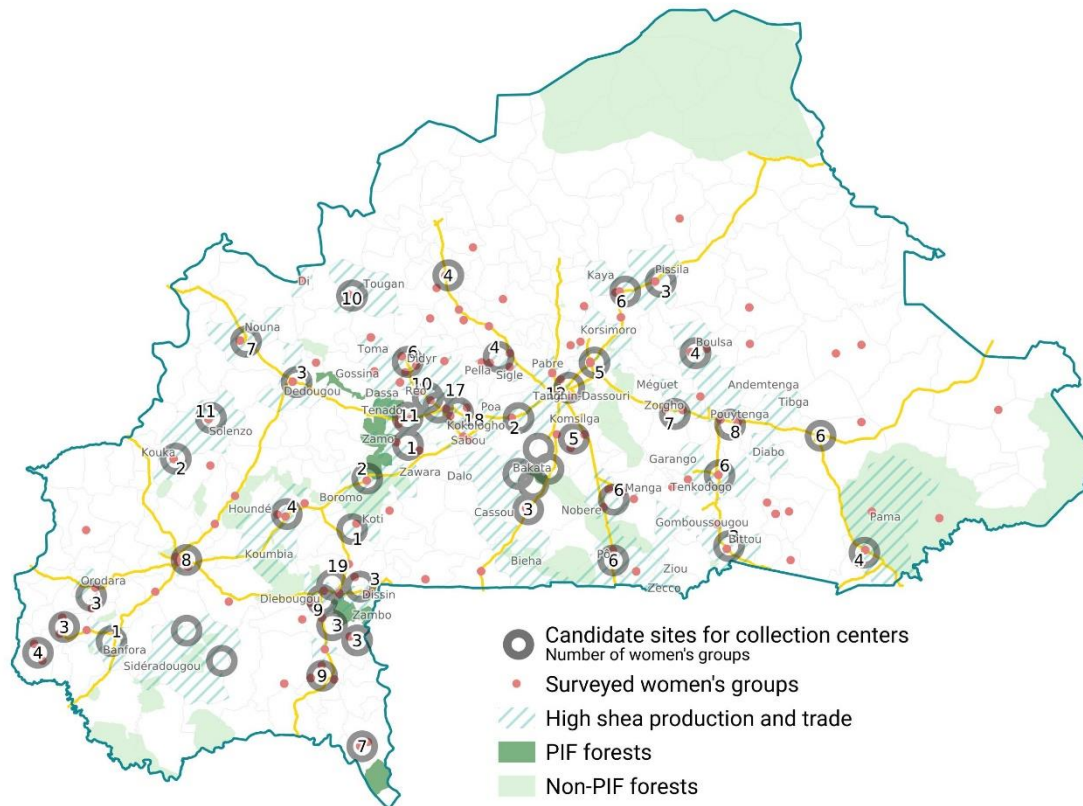
2. THE MARKET

Market: Selected examples of personal care companies engaged by Serious Shea

Name of Company	Business Function	Requests and Requirements	Volumes Purchased per Year	Indicative Unrefined Shea Butter Prices / kg
Herbal Essentials 	Brand	Uses external sourcing agent to purchase natural shea butter	<50t natural raw shea butter	TBC
Lamotte 	Processor; sale to formulators	Offer premium on natural carbon neutral shea butter	350 tonnes unrefined butter	TBC
Interfat Natural Oils 	Formulator & Distributor	32°C melting point. Marseille distribution requested	<50t natural raw shea butter	€3 in last harvest
Hallstar 	Formulator & Distributor	Smaller clients: Sustainability focus ; Finding sustainability marketing opportunities	>140t	N/A
Textron 	Formulator	Packaging in metal drums for reuse not plastic drums	>70t	€2.40-€2.50/kg
Fourchem 	Supplier of natural ingredients	Focus on price. Buys from Savannah Fruits	<50t	€2.40-€2.50/kg
Kerfoot 	Cosmetic formulator and ingredients supplier	Focus on natural ingredients, sustainability, women with activities in production areas	>100t	TBC
Adina 	Distributor	Sustainability focus - nut to end product traceability increasingly demanded.	TBC	TBC
L'Occitane 	Retailer	Women & Fair Trade Focus	>750 tons of shea butter	\$2.76/kgf natural, fair trade (\$3.45 organic)
Berg + Schmidt 	Processor; sale to formulators	New focus on Focus on organic products for cosmetics markets; Place orders Q3 each year	>1000 tonnes	TBC

3. IMPACT

Social impact: Serious Shea is moving production and processing into the local communities, improving the livelihoods of over 10,000 women producers

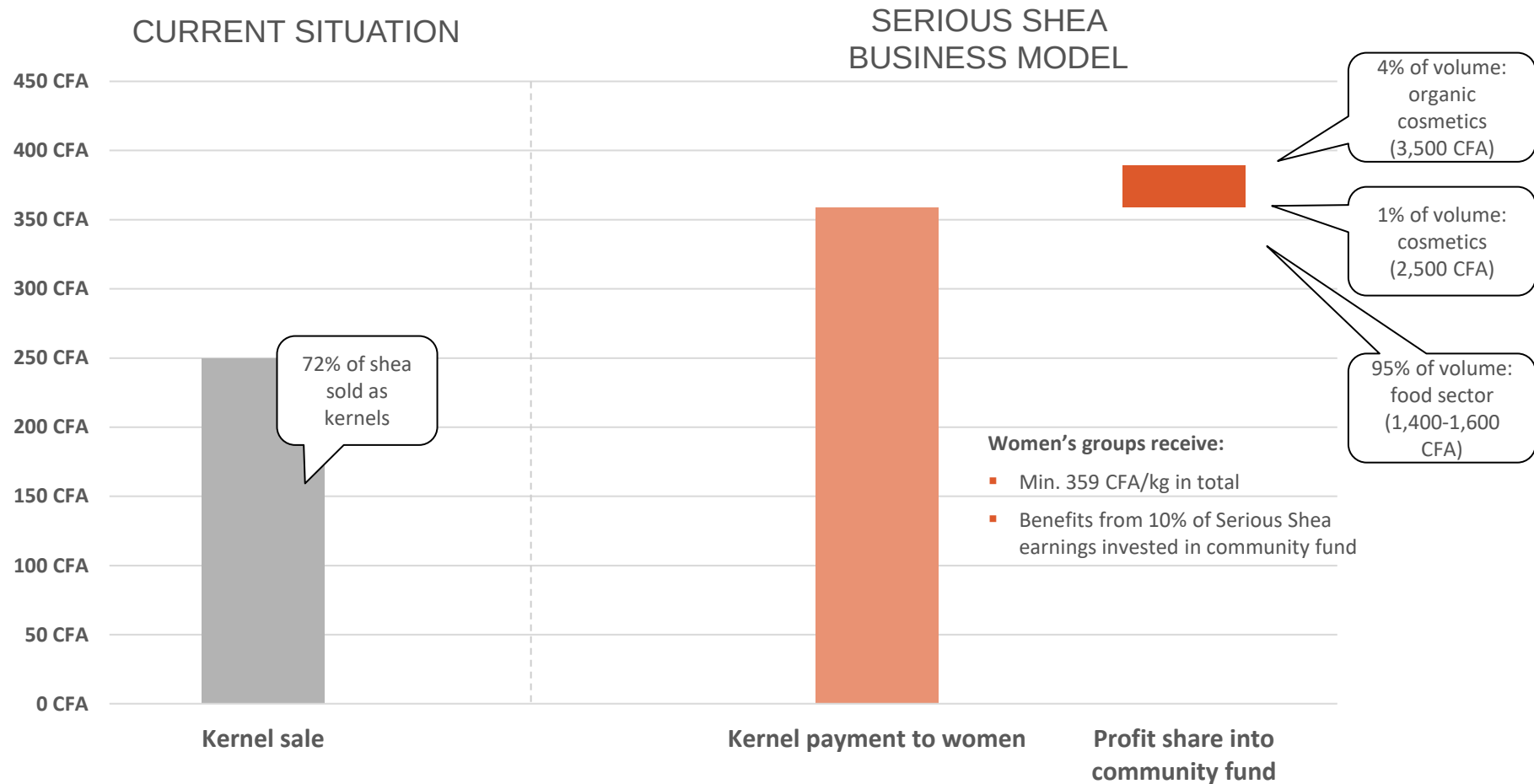


- Serious Shea will be operating multiple collection sites and large processing centres embedded close to women processing groups
- Working with survey groups in-country, the team has already identified and engaged >300 women's groups (>4,000 women) to understand current practices, and the appetite for new technology and methods
- Serious Shea will revolutionise the Burkina Faso shea sector:
 - Transform the lives of over 10,000 participating women producers, providing clean processing technology and higher incomes
 - Each producer is expected to earn 1.5x higher income for the butter that they produce

Site locations: Serious Shea has engaged >300 women's groups across the country to understand their preferences and confirm their support. The team has also identified the above sites for nut collection based on detailed mapping, logistics, forestry and social assessments

3. IMPACT

The margins for production have to ensure a good price for women while maintaining profitability for the overall business



3. IMPACT

Environmental impact: Reducing deforestation and cutting carbon emissions



- The large consumption of wood for fuel in shea butter processing leads to significant greenhouse gas emissions. In traditional methods, non-renewable biomass is inefficiently consumed in boiling nuts, denaturing & de-husking, roasting, heating water for kneading, and boiling fat to dry and clean, and accounts for $\sim 25\text{kgCO}_2\text{e/kg}$ butter at farm.
 - *Shea is an endangered species protected by CITES and the national forest code. It continues to be exploited for biomass, as a fuel for heating and for the production of coal*
 - *In Burkina Faso, between 1990 and 2010, the total forest area has decreased by approx. 50,000 km², representing a loss of 17.5%*
- Both the deforestation and carbon emissions could be reversed and negated respectively through the implementation of the Eco-Processing Centres, as they will minimise the need for fuelwood, and enable the protection and reforestation of forest populations. Each centre is expected to save 15,000tCO₂e/year of GHG emissions and drive widespread reforestation

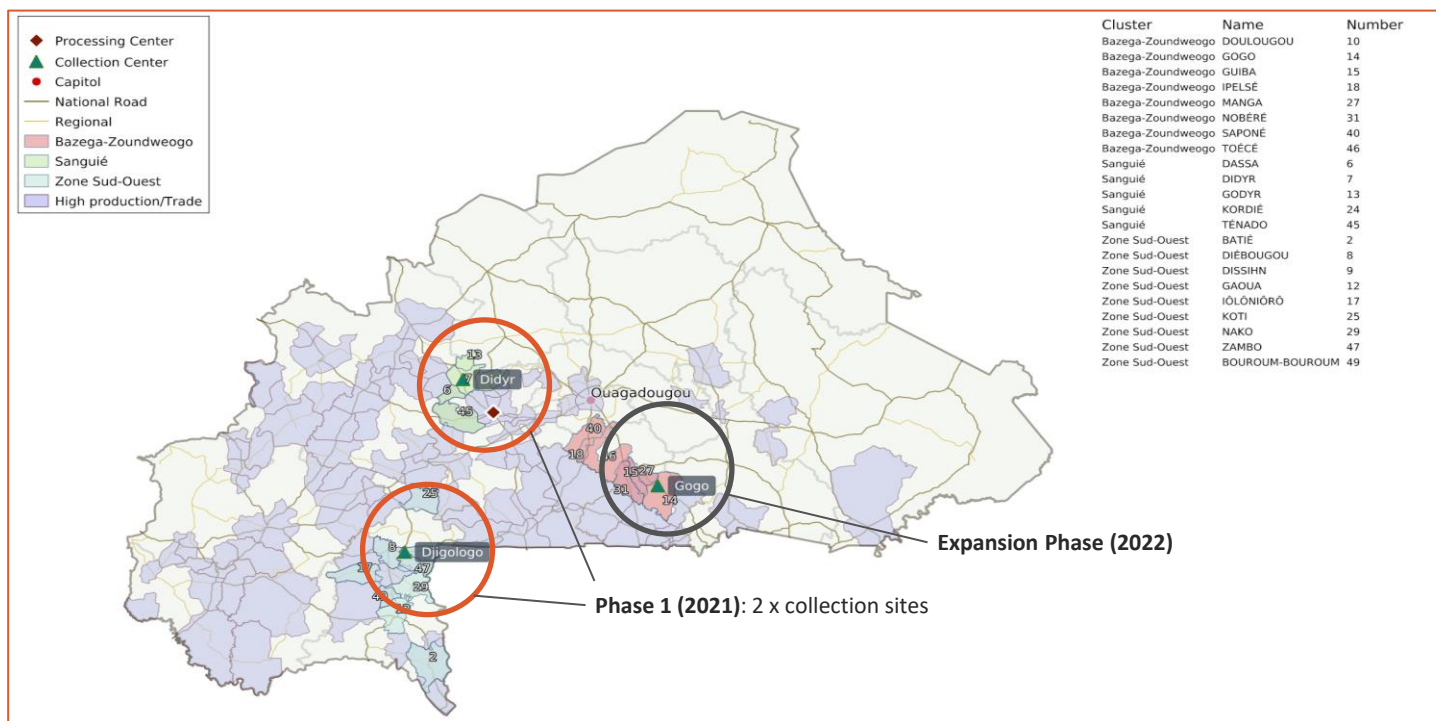


5. PHASE 1

As Phase 1, the business will begin implementing the first processing centre in 2021

2021 Plan: Phase 1

1. Construct 8 x collection sites (starting in Didyr, Sanguié Province and Djigologo, Loba Province, Sud-Ouest Region) and 4 x processing centre (in Koudougou, Boulkiemdé Province) in 2021
2. Collect ~6,000mt of dried kernels, and produce and export ~2,400mt of shea butter in 2021/22 (Revenue: >US\$9m)
3. Create direct employment for 1,280 staff (90% women) and benefit 10,000s of women processors by offering them ~1.5x higher prices for the dried kernels and 10% of the business profits



6. CONTACT

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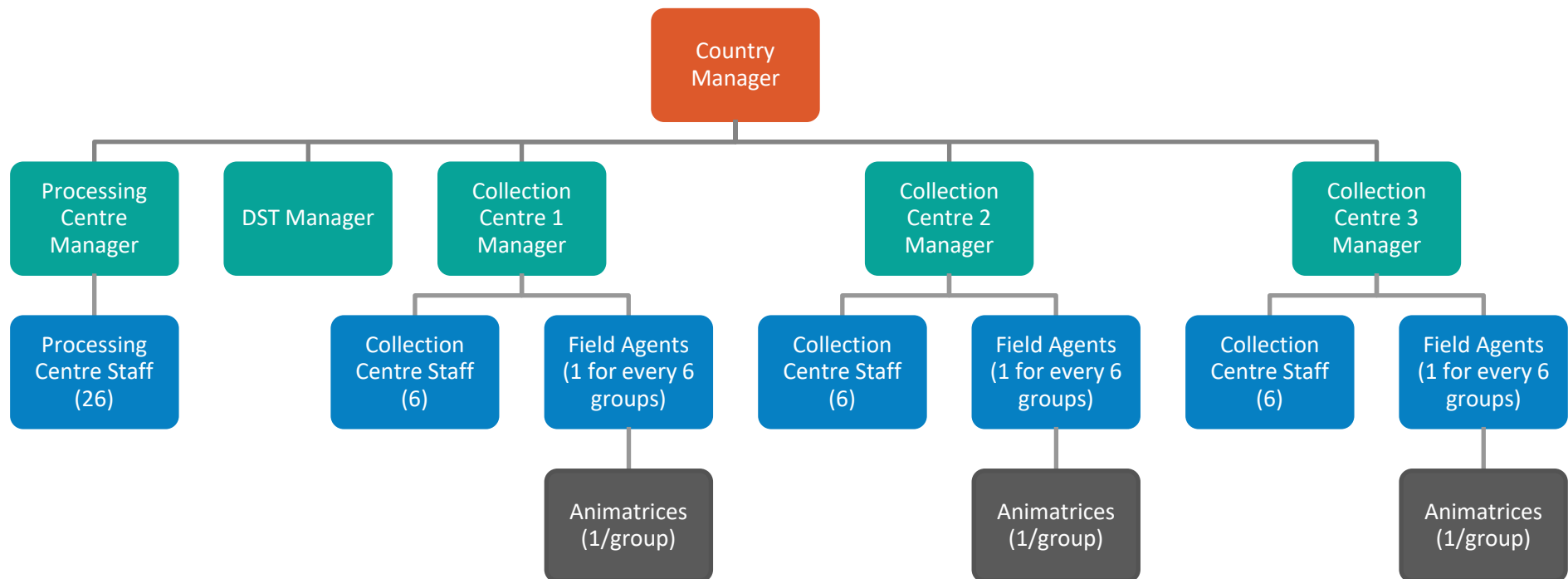
wkwende@agritechfaso.com

+33 (0)6 14 45 39 63

ANNEX: SERIOUS SHEA ORGANIGRAM

Serious Shea will be managed on-ground by a Country Manager, who will oversee the processing centre and collection centres

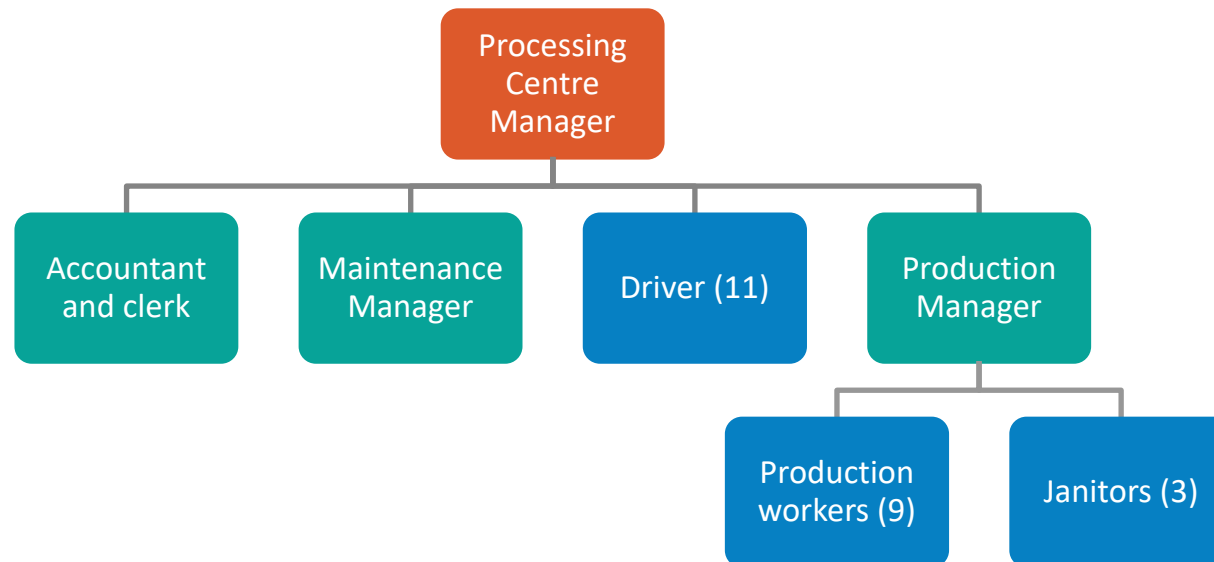
- Each collection centre will be overseen by a Centre Manager, who will oversee:
 - Centre staff (15 for the processing centre and 11 for each collection centre)
 - Associated field staff (1 Field Agent for each 6 collector groups and 1 Animatrice for each collector group)



ANNEX: ORGANIGRAM - PROCESSING CENTRE

The Processing Centre will be managed by a Centre Manager, who will oversee 11 centre staff and associated field staff

- Key processing centre staff include an Accountant, a Maintenance Manager, a Clerk and a Production Leader



Centre Manager

- Manages centre staff & Union/Group Agents
- Aggregates Union/Group projections and targets
- Prints QR codes for distribution
- Oversees sourcing volume against targets

Production Manager

- Manages 4 production staff and 3 janitorial staff

Clerk

- Manage purchasing and warehousing
- Responsible for dispatch and organizing logistics
- **Accountant**
- Daily financial reporting, payroll, audit requirements
- Financial analysis

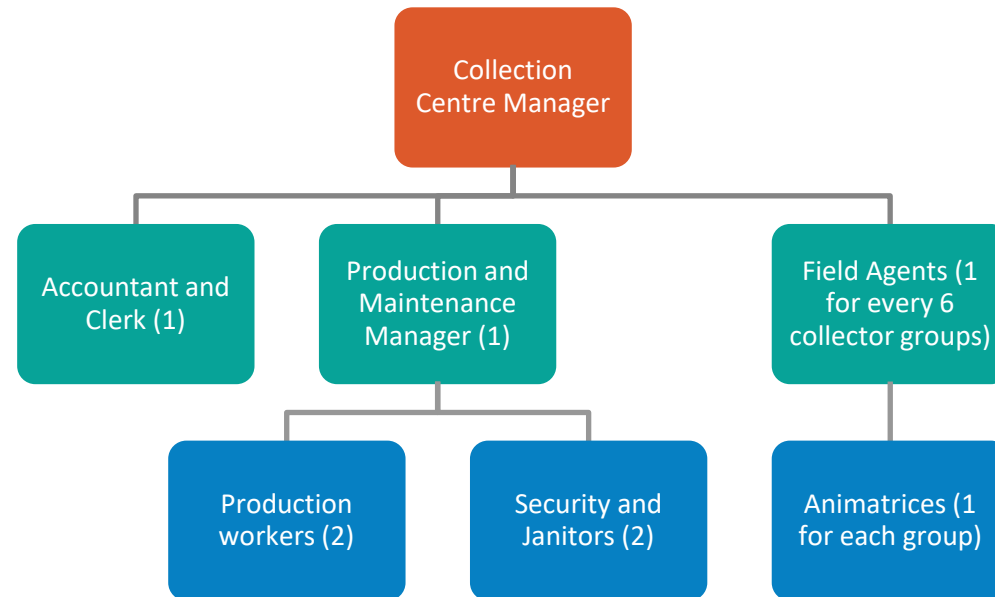
Maintenance Manager

- Install, repair, upkeep the equipment

ANNEX: ORGANIGRAM - COLLECTION CENTRE

Each Collection Centre will be managed by a Centre Manager, who will oversee centre staff plus field staff

- Field Agents: 1 for every 6 collector groups
- Animatrices: 1 for each group



Animatrices

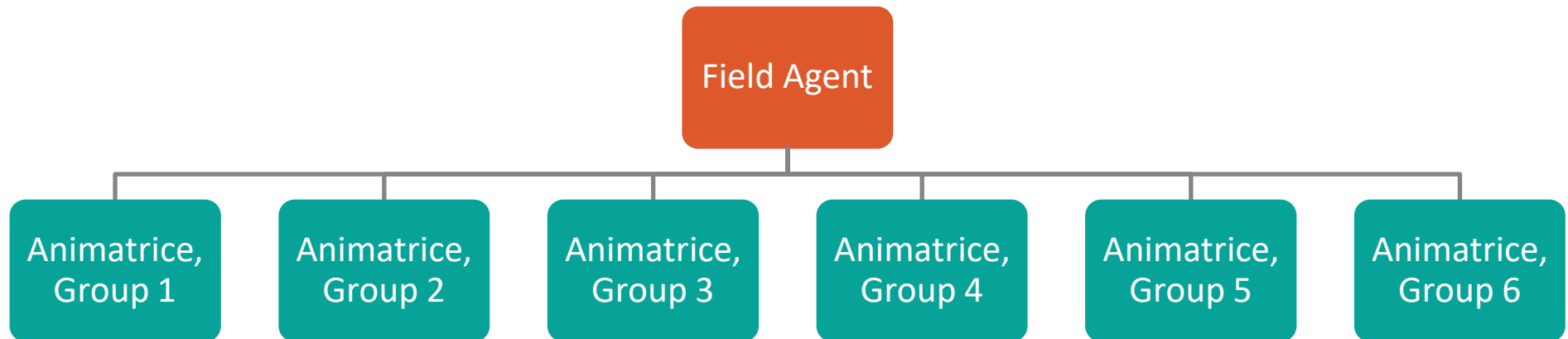
- Known and trusted by the communities/groups
- Main field trainers who provide day to day monitoring of collector and group activities
- Literate, but not be able to enter data via Smartphone

Field Agents

- Manage Animatrices & oversee buying functions of groups
- Oversee the entry and validation of key field data via Smartphone

ANNEX: KEY ROLES & RESPONSIBILITIES OF FIELD STAFF

Each Field Agent Covers up to 6 collector groups, each of which is overseen by an Animatrice



Field Agent

Coverage

- In Unions requiring multiple Agents, 1 Senior Agent should manage the other agents, including their annual budget for capacity building.
- In Groups not belonging to Unions, 1 Agent can cover up to 6 groups, depending on geographic spread.

Key Responsibilities

- Manage sourcing targets and projections for their assigned groups (awarded bonuses accordingly)
- Train and oversee group purchasing agent; monitor purchasing data via the DST
- During the buying season, visit each group weekly to quality control and collect purchases

Animatrice

Coverage

- At least one Animatrice per group (or 1 per community, if groups' members are spread out)

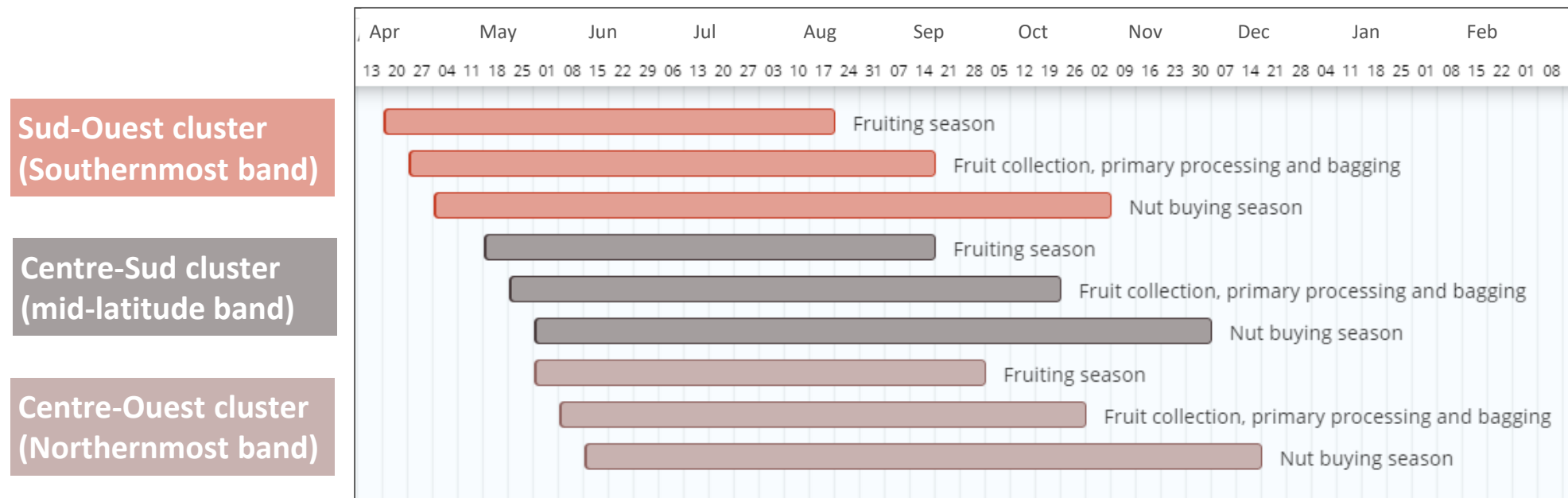
Key Responsibilities

- Assist the Field Agent in communications and logistics with the group
- Conduct community sensitization
- Coordinate registration of collectors (via Field Agents)
- Distribute bags and printed QR codes to collectors
- Conducts collector registration, training, & quality monitoring

ANNEX: FRUITING AND BUYING SEASONS

The Serious Shea buying season will be staggered based on lateral climatic bands

- Shea fruiting and therefore the buying season begins earlier in the south, and can be characterized into 3 lateral bands.
- Implications for the Serious Shea pilot clusters are as follows:



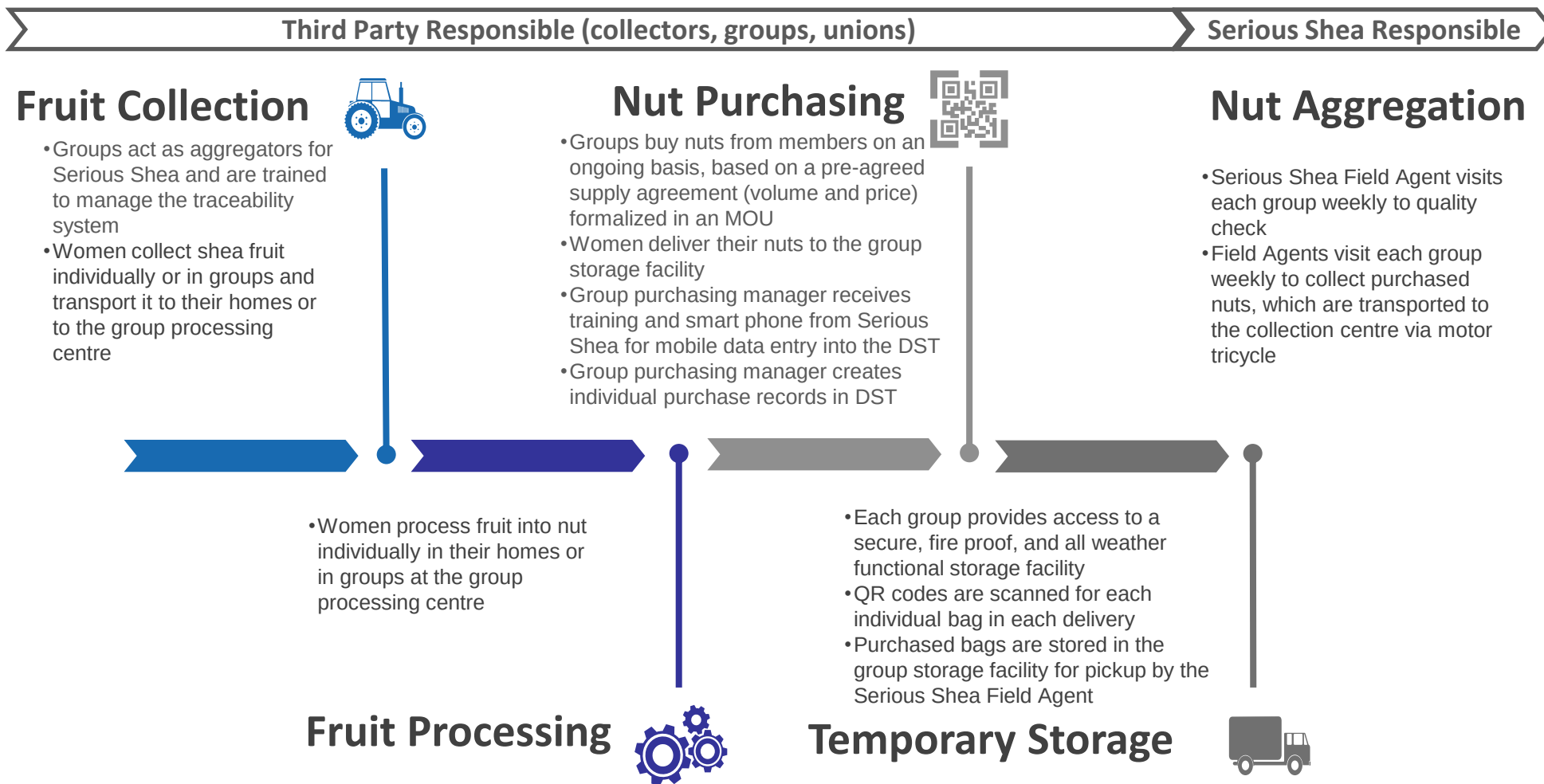
Data source: Brigitte Bastide, unpublished data from INCO-Innovkar project, 2008-2010

*The Sud-Ouest cluster is slightly north of the southernmost band, and the buying season may therefore come slightly later

**The Centre-Ouest/Sanguie cluster falls between the northernmost and mid-latitude bands, and the buying season may therefore come slightly earlier.

ANNEX: HARVEST TO COLLECTION SITE

Groups will purchase and aggregate shea nut for periodic pickup by the Field Agents via motor tricycle



ANNEX: COLLECTION CENTER TO PROCESSING CENTRE

Shea nut will be delivered to the collection centre for processing into kernel, then stored until being transported to the processing centre

Serious Shea Responsible

Nut Reception



- Field Agents deliver by motor tricycle shea nut to the collection centres from the group storage
- Quality control at collection centres performed by the production manager before accepting the material purchased from the field

Kernel Storage



- Dried kernels are bagged, bar-coded and batched
- Bar codes are scanned
- Bags are entered into storage

Kernel Reception

- Quality control at processing centre performed by the production manager before accepting material from the collection centre
- Batches of dried kernel are received by the processing centre clerk
- Clerk scans all barcodes
- Bags are entered into storage prior to processing

- Shea nuts are dried, graded, and cleaned
- Nut is heated, boiled, de-shelled, and dried by collection centre production staff

Nut Processing



- Bags are backloaded into trucks and driven by independent operators to the processing centre

Kernel Transport



There will be 11 key operational functions

1 Identification of New Groups/Unions in the Clusters

- Prospection, identification, and networking with additional women groups/unions (those not already identified by TechnoServe & Sensonomic surveys) in the clusters to reach volume targets

2 Sensitisation of Unions and Groups

- Introduction of collector organisations to Serious Shea's business model, including pricing
- Begin building relationships, gauge interest in participation, begin estimating potential supply volume

3 Signing of MOUs

- Formalization of relationships with supplier collector organisations
- Memoranda of Understanding to formalize the buying agreement between the organisation and Serious Shea
- Should define projected volume to be purchased, the form of shea (nut), estimated delivery date, price and payment terms

4 Collector Registration

- Explanation of Serious Shea's business model and pricing terms to individual collectors
- Individual consent of each collector to become a Serious Shea supplier
- Creation of a unique collector ID and record for each woman, to enable traceability

5 Arrangement of Field Storage Facilities

- Contracting of existing storage facilities to ensure adequate storage for nut purchased
- Formalized via a rental agreement that includes storage capacity provided, pricing, payment terms, maintenance conditions and management modality

ANNEX: KEY OPERATIONAL FUNCTIONS (CONTINUED)

6 Development of Volume Projections

- Enter official volume projections into the DST via mobile application
- Data triangulation through projections at both the individual and group levels, for cross-checking
- Enable monitoring against expected deliveries from individual collections
- Enable identification and reward of high performing collectors

7 Registration of Village Forests/Parklands

- Register access to village forests and parklands through local government officials

8 Union & Group Leadership & Management Training

- Collector organisations must operate the traceability system on an ongoing basis, including managing of purchasing and warehousing functions via mobile data entry into the DST
- Training to be provided on operation of the DST and related equipment, as well as Serious Shea standard operating procedures
- Additional training to be provided on group dynamics, membership management and (eventually) certification norms and practices

9 Printing & Distribution of QR Codes

- QR codes are generated for each individual collector, based on projected volume for that supplier, and connected to the supplier's collector ID
- QR codes are printed using a special printer at the processing centre, then distributed to the collection centre, then to the Field Agents for distribution to the Animatrices, for distribution to the collectors
- *See Annex and DST guidance documents for detailed instructions on generating and applying QR codes*

10 Collector Training on Quality & Traceability

- Ongoing training on shea best practices, based on Global Shea Alliance training manual
- Emphasis on environmental responsibility, traceability and respect of Serious Shea supply terms

11 Buying Season

- Serious Shea field agents visit participating groups/unions once a week to collect purchase volumes

ANNEX: TRANSPORT PROCEDURE

Purchased motor tricycles will transport nuts; contracted backloaded trucks will transport kernel and butter

- Nut purchases aggregated by groups and stored in their warehouses
- Field Agents equipped with moto tricycles visit each group warehouse weekly during buying season
- Field Agents monitor group purchases via the DST
- Field Agents quality check stored bags, scan barcodes of accepted bags and transport purchases to Collection Centre

- Backloading of 40T trucks via the same “Station Managers”
- There are fixed routes with relatively standard pricing per 100kg from Ouaga to Tema
- Greater possibility for negotiation of annual fixed pricing via fixed contracts
- *Price benchmark (XOF 21,000 per MT, to be confirmed by Karikis)*



- Backloading of 8T Kia trucks via local “Station Managers” who manage fleets of trucks
- Station Managers program onloading and offloading of cargo on various trucks in the fleet
- Pricing is based on distance and weight and are all inclusive of driver, fuel and road toll costs
- Agreements tend to be informal, but fixed pricing may be formalized in annual agreements, once volumes are better known
- *Price benchmark (Tenado to Ouaga, 125km, FCFA 1,000 per 85/kg bag) to be confirmed by Karikis*

ANNEX: TRANSPORT TIMING

Transport of nut will be weekly, whereas transport of kernel and butter will be based on critical volumes

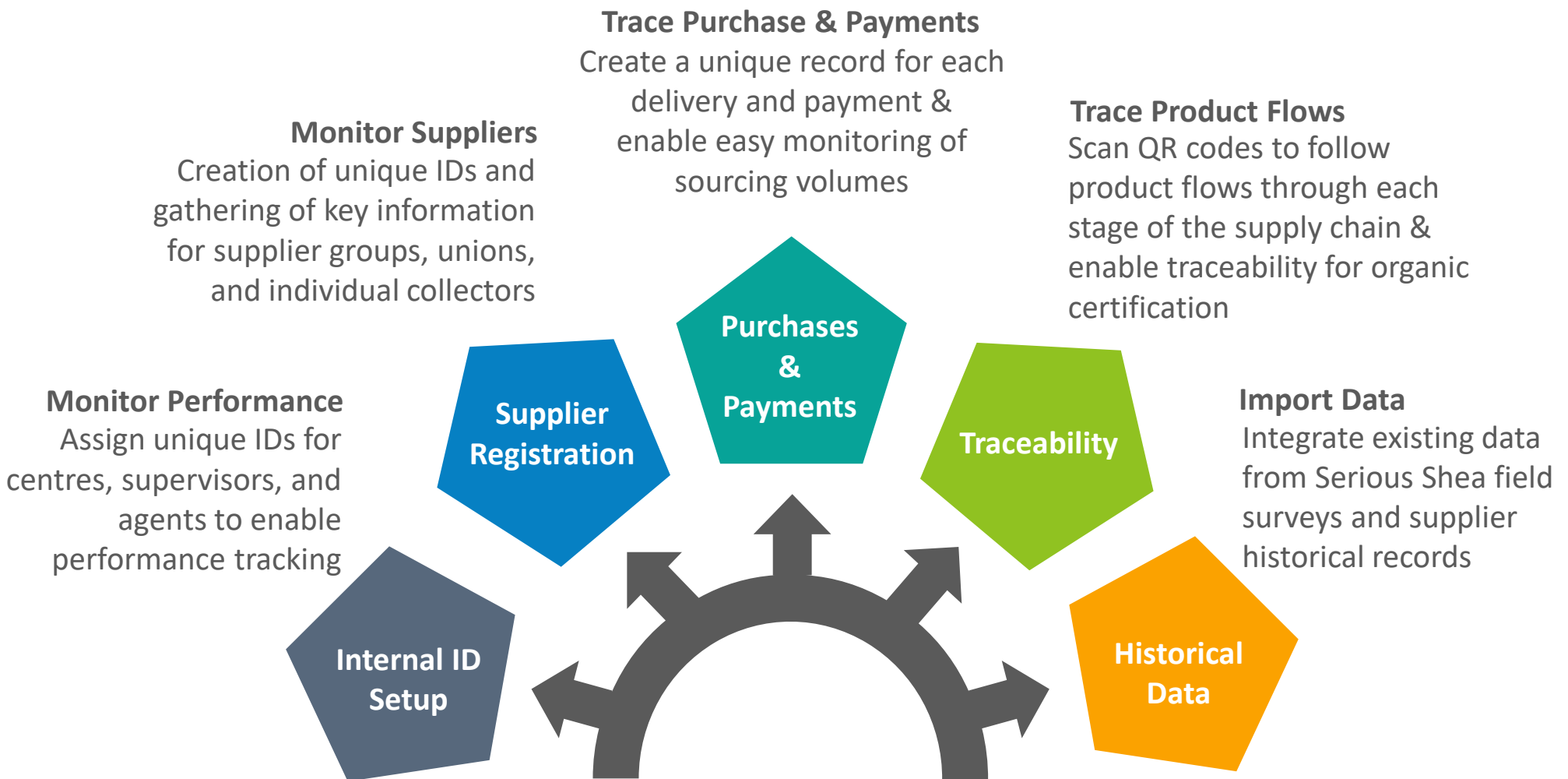
- Weekly, or as needed
- Additional pickups may be programmed based on Field Agent monitoring of group purchases via the DST
- Based on how quickly Serious Shea can fill a truck (max 40T)
- Assuming 740MT of butter annually (2,000 MT nut, transformation ratio of 37%), Serious Shea will need to transport at least 19 truck loads.



- Based on how quickly Serious Shea can fill a truck (max 40 MT)
- Assuming 1,500 MT of kernel annually (2,000 MT nut, transformation ratio of 75%), and thus ~500 MT of kernel per Collection Centre, Serious Shea will need to transport at least 13 truck loads per centre, or roughly 1 truck per week throughout the buying season

ANNEX: DECISION SUPPORT TOOL (DST) UTILITY

The DST will monitor performance and suppliers, trade purchases, payments and product flows, and allow for data imports



Key data will be entered via mobile app to create each purchase record

